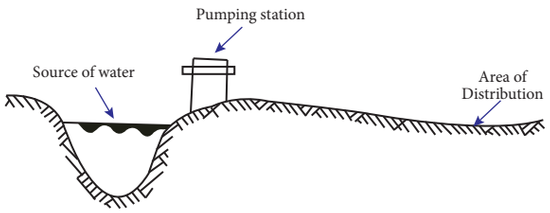
Demerits

i. Source of water supply should be higher than the area served.

ii. During fire-fighting, it may require motor pumping to develop required pressure.

b. Pumping system

This system is followed when the source of water supply is lower than the area served or the ground between the source and the distribution area is undulating. In this system, treated water is pumped directly into the water distribution mains. Water flows under hydraulic pressure.



Merits

i. The source of supply can be at any level.

ii. No pumping during non-supply

Demerits

i. Requires high lift pumps.

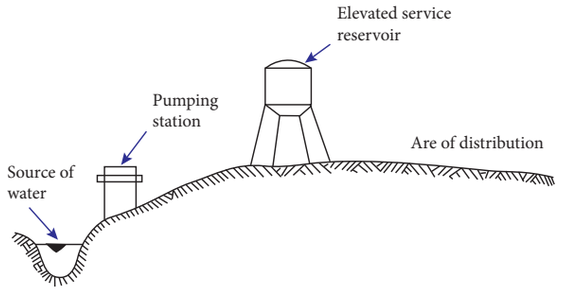
ii. Requires constant attention at pumping stations.

iii. Unreliable as it depends entirely on power supply.

iv. Pressure in mains varies as the variation in consumption.

c. Combined pumping and gravity system

In this system water is supplied by both gravity and pumping. When the demand is less than the rate of pumping, the excess water is stored in the reservoir. When the demand is more than the rate of pumping, the water from the reservoir is supplied for consumption.



Merits

i. Most economical, efficient and reliable.

ii. Pumping can be done at uniform rate.

iii. Pumps can operate at their capacities which results in higher efficiency and economy in operation.

iv. Stored water can be used at all emergencies.

4.7.2 Systems of water supply

Based on the frequency of water supply, the following water supply systems may be followed.

a. Continuous system

b. Intermittent system

a. Continuous system

In continuous system, treated water is supplied to the consumers throughout the day.

Merits

i. Most ideal.

ii. No separate storage by the consumers is required.

iii. Adequate supply is always available for fire-fighting.

iv. Wastage is minimum.

Demerits

i. The water supply source should have adequate quantity.

ii. Wastage will be more if the consumers do not have civic sense.

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